GEOMETRIC GRAPH RESEARCH / CONSOLIDATION

# What survives the audit?

## A fixed-interior moment theorem, a frozen boundary-density test, and a competing model for the fan evidence

16 September 2026

**Main analytical advance.** An independent-group enclosure argument gives a uniform stretched-exponential bound on the rescaled outer radius of the cell containing a *fixed interior point*. This supplies the uniform integrability missing from the earlier local-limit report and yields convergence of all polynomial area and perimeter moments in the stated setting.

**Main correction.** Matching inward and tangential gap distributions, even together with adjacent inward-gap dependence, does not identify the proposed two-dimensional fan process. An explicit random rectangular grid matches every one of those limiting observables. A perpendicular two-ray event sharply separates the models: approximately 4.85% versus 17.40%. These are model predictions, not newly measured graph probabilities.

**One precommitted numerical test.** A positive nonuniform city density with a 9:1 boundary contrast was tested against frozen predictions. The experiment used 1,600 independent graphs and 86,686 exact hull-adjacent triangles. A further 80 small graphs supplied independent polygonization validation. Finite-size bias was retained, not fitted away.

**Status.** Proofs below are supplied for review, not independently peer reviewed or machine verified. The boundary coefficient remains conditional on a localization step. The complete-graph fan-process limit remains a conjectural connection. Established Poisson, Cox, and random-polytope results are credited. A scoped public-web search cannot certify novelty.

**Scope of this archive.** New code, frozen plan, per-graph results, claim ledger, source map, search log, checksums, and editable report source. Earlier archives and third-party papers are not duplicated.

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# 1 Audit decisions: fewer headline claims, clearer evidence

The previous phase mixed exact finite geometry, classical limit-model formulas, numerical checks, and incomplete transfers from a proposed limiting model to complete graphs. These are not interchangeable. The audit reads the earlier source reports and code as *inputs*; the previously reported large simulations are not represented as rerun here.

| Claim | Audit status | Consequence |
|---|---|---|
| Fixed-interior Poisson-line limit | Can be obtained from an established functional Poisson theorem after explicit geometric checks | Present as an application with proof, not as a new general Poisson theorem. |
| Fixed-interior area/perimeter moment limits | Proof completed below using independent groups | This closes a specific earlier gap; it is not a numerical improvement. |
| Boundary-face area constant | Convergent integral in the parabolic model; transfer to finite graphs still conditional | Keep the constant and experiments, but label the asymptotic claim with its unresolved localization requirement. |
| Intermediate Cox fan process | Exact formulas within the specified model; full complete-graph process limit not proved | Keep selected-observable evidence separate from a process convergence theorem. |
| Novelty of the package | Not established | Identify antecedents and unresolved priority rather than assigning a probability of novelty. |

A simulation interval measures Monte Carlo uncertainty under a specified implementation. It does not measure the probability that a conjecture is true, include unbounded asymptotic bias, or account for every hypothesis selected during earlier exploration.

# 2 The fixed-interior theorem

## 2.1 Setting and conventions

Let $D \subset \mathbb{R}^2$ be a compact convex body with nonempty interior and area $S$. Let $X_1, \dots, X_n$ be iid uniform points in $D$, and draw every finite segment $[X_i, X_j]$, $i < j$. Fix $x \in \operatorname{int} D$ deterministically, independently of the sample.

When $x$ is inside the random convex hull, let $C_n(x)$ be the closure of its containing bounded face. When it is outside, set $C_n(x) = \{x\}$ for the purposes of the following observables:

$$A_n = \operatorname{area}(C_n), \qquad P_n = \operatorname{perimeter}(C_n), \qquad R_n = \sup_{y \in C_n} \|y - x\|.$$

All three are zero in the outside-hull case. $R_n$ is an *outer radius about the probe*, not a maximum inscribed-circle radius. Events where $x$ lies exactly on an edge have probability zero.

Bounded faces are convex almost surely. At a proper crossing, all sectors have angle below $\pi$. At an interior city, the other cities surround it, so no consecutive ray gap can exceed $\pi$; at a hull city, the bounded-face sectors lie within its convex-hull angle. The planarized complete graph has no articulation creating a repeated boundary walk. Hence bounded-face boundary cycles have no reflex corner. In particular,

$$A_n \le \pi R_n^2, \qquad P_n \le 2\pi R_n. \tag{1}$$

Parameterize unoriented lines by direction $\theta \in [0, \pi)$ and signed perpendicular offset $p$. The usual identification at the endpoints of the direction interval makes this the locally compact space of unoriented lines. Put $u_\theta = (\cos\theta, \sin\theta)$. If $a_\theta, b_\theta$ are the distances from $x$ to $\partial D$ along $u_\theta, -u_\theta$, define

$$\lambda_x(\theta) = \frac{a_\theta b_\theta (a_\theta + b_\theta)}{2S^2}. \tag{2}$$

Let $\mathcal{L}_x$ be the stationary Poisson line process of intensity $\lambda_x(\theta)\,\mathrm{d}p\,\mathrm{d}\theta$, and let $Z_x$ be its zero cell, the cell containing the origin.

**Theorem 1** (Fixed-interior local and moment limits)**.** *In this setting, the rescaled containing cell $n^2(C_n(x) - x)$ converges in distribution to $Z_x$ in the Hausdorff topology on compact convex sets. There are constants $C, c > 0$, depending on $D, x$ but not $n, t$, such that*

$$\boxed{\mathbb{P}(n^2 R_n > t) \le Ce^{-c\sqrt{t}}, \qquad t \ge 0, \quad n \ge 3.} \tag{3}$$

*Consequently, for every real $q > 0$,*

$$\lim_{n\to\infty} \mathbb{E}[(n^4 A_n)^q] = \mathbb{E}[\mathrm{area}(Z_x)^q], \tag{4}$$

$$\lim_{n\to\infty} \mathbb{E}[(n^2 P_n)^q] = \mathbb{E}[\mathrm{perimeter}(Z_x)^q]. \tag{5}$$

*The analogous assertion holds for the outer radius. This is a fixed-probe result; no uniformity for probes approaching $\partial D$ is asserted.*

The local point-process component is an application of Decreusefond–Schulte–Thäle [1, Theorem 3.1]. The enclosure lemma in Section 3 supplies the additional tail control. The proof below is written out to make that division of labor auditable.

### 2.2 Mean measure: the finite segments must actually reach the probe

Write $\varepsilon = n^{-2}$. For each unordered pair, retain its supporting line only if its orthogonal projection of $x$ falls *within the finite segment*, and its signed distance to $x$ has magnitude at most $R\varepsilon$. Record its direction and offset divided by $\varepsilon$. The resulting finite point process on the line window is a symmetric two-point statistic of the binomial city process.

The line-coordinate map for two ordered endpoints has Jacobian equal to their separation along the line. Its two straddling orders cancel the factor $1/2$ from unordered pairs. The limiting mean density is therefore

$$\int_0^{a_\theta} \int_0^{b_\theta} \frac{s+t}{S^2}\,\mathrm{d}s\,\mathrm{d}t = \lambda_x(\theta).$$

More precisely, the prelimit mean density in the rescaled line window is

$$(1 - 1/n)\lambda_{x+\varepsilon p v_\theta}(\theta),$$

where $v_\theta$ is a perpendicular unit vector. Chord lengths through interior points vary continuously. They are bounded on the fixed window and the limiting intensity is positive in every direction. Dominated convergence gives total-variation convergence of the mean measures.

This endpoint restriction is essential. Counting all infinite supporting lines would include pairs on the same side whose finite segment never visits the probe. The probability that any city lies in a ball of physical radius $O(\varepsilon)$ is $O(n\varepsilon^2) = O(n^{-3})$. Outside that event, traces meeting a smaller observation ball traverse it completely, so the restricted line process and actual segment traces agree there.

### 2.3 Shared-label dependence: checking the published theorem

For a city fixed at $z$, write $p_\varepsilon(z)$ for the probability that its segment to one other city hits a fixed-radius $\varepsilon$-scaled ball about $x$. A cone-angle bound gives

$$p_\varepsilon(z) \le C \min\{1, \varepsilon/\|z-x\|\}.$$

The density is bounded and $D$ is compact. Splitting into distances below $\varepsilon$ and annuli above it gives

$$\begin{aligned}\int_D p_\varepsilon(z)^2 \frac{\mathrm{d}z}{S} &\le C\left[\varepsilon^2 + \int_\varepsilon^{\operatorname{diam} D} (\varepsilon/r)^2 r \,\mathrm{d}r\right] \\ &\le C\varepsilon^2(1+|\log\varepsilon|).\end{aligned}$$

Two pair statistics with disjoint city labels are independent. For two edges sharing a city, conditional independence of their other endpoints gives exactly the overlap integral above. In the notation of [1], the order-two overlap functional is bounded by

$$n^3 \int p_\varepsilon(z)^2 \frac{\mathrm{d}z}{S} = O(\log n/n).$$

The remaining binomial correction in their theorem is $O(1/n)$ since the expected line-window count is bounded. Thus their bound yields

$$d_{\mathrm{KR}}(\Xi_{n,R}, \mathcal{L}_{x,R}) \le d_{\mathrm{TV}}(\mu_{n,R}, \mu_R) + O(\log n/n) \longrightarrow 0.$$

The displayed $O(\log n/n)$ controls the dependence term; a quantitative bound on the separate mean-measure error is not silently assumed. This is a functional point-process theorem, not merely convergence of the number of cuts.

### 2.4 From line windows to the containing cell

The limiting process has positive intensity in enough direction intervals to enclose the origin almost surely. Its zero cell is bounded, and finite-window configurations are nondegenerate almost surely: no line through the origin, no relevant coincident lines, and no triple crossing. Choose a large window in which the zero cell is enclosed with probability close to one. Small perturbations of the finitely many enclosing lines preserve enclosure in a slightly enlarged window and vary the cell continuously.

The line-window convergence and the vanishing city-endpoint event transfer this enclosure and continuity to the finite graph. Increasing the window gives Hausdorff convergence of the containing cells and tightness of $n^2 R_n$. The outside-hull convention causes no asymptotic change because an enclosing local cycle guarantees hull containment, whose probability tends to one.

This establishes distributional convergence, but on its own would *not* establish the expectation limits. The next argument closes that gap.

## 3 Independent groups control the rare giant cells

**Lemma 2** (Enclosure amplification)**.** *Suppose that, for all $m \ge m_0$, an induced complete graph on $m$ iid cities has probability at least $p > 0$ of enclosing $x$ in a bounded cell of outer radius at most $R_*/m^2$. Then the full $n$-city graph satisfies* (3)*, with constants depending on $p, R_*, m_0, \operatorname{diam} D$.*

*Proof.* Partition the city labels into $q = \lfloor n/m \rfloor$ disjoint groups of $m$ and ignore leftovers. Their induced graphs are independent. If even one group's cell encloses $x$ inside a ball, adding all cross-group edges cannot enlarge that containing cell. Therefore, whenever $R_*/m^2 \le t/n^2$,

$$\mathbb{P}(n^2 R_n > t) \le (1-p)^{\lfloor n/m \rfloor}. \tag{6}$$

For sufficiently large $t$, choose

$$m = \max\{m_0, \lceil n\sqrt{R_*/t} \rceil\}.$$

In the nonsaturated case, rounding only changes constants and $q \ge c_1\sqrt{t}$. In the saturated case $m = m_0$, we have $q \ge c_2 n$ for $n \ge 2m_0$. The event is impossible when $t > \operatorname{diam}(D)n^2$. Otherwise $n \ge \sqrt{t/\operatorname{diam} D}$, so again $q \ge c_3\sqrt{t}$. Equation (6) now yields $Ce^{-c\sqrt{t}}$. Bounded $t$ and the finitely many $n < 2m_0$ are absorbed by increasing $C$. □

The lemma's assumption follows from the local cell limit: choose a continuity radius $R_*$ so that the limiting zero cell lies strictly within that radius with probability above, say, 3/4. For all sufficiently large $m$, the corresponding *bounded enclosure event* has probability at least 1/2. It is important to use a successful bounded enclosure, not simply the numerical value $R_m = 0$ assigned outside the hull.

*Completion of Theorem 1.* Lemma 2 proves the uniform tail estimate. Integration of that tail shows

$$\sup_{n \ge 3} \mathbb{E}[(n^2 R_n)^s] < \infty \quad \text{for every } s > 0.$$

For uniform integrability at any desired power, use a strictly higher power. Equation (1) transfers these bounds to $(n^4 A_n)^q$ and $(n^2 P_n)^q$. Area and perimeter are continuous on compact convex sets under Hausdorff convergence. Distributional convergence plus uniform integrability therefore yields (4)–(5). □

### 3.1 Why this matters, and what it does not imply

The enclosure argument uses completeness in an essential way: the full graph contains every independent subgroup graph. A sparse graph need not inherit a successful subgroup's enclosing cycle.

The constants depend on the fixed probe's distance from the boundary and on the domain. They are not asserted uniform as the probe moves. Consequently this result does not contradict the earlier boundary-dominated spatial averages. It also does not establish convergence of side-count moments, a sharp finite-$n$ rate for the moments, or an asymptotic law for the globally largest cell.

The mean convergence also holds conditional on the probe lying inside the random hull: under the zero convention the unconditional moment divided by that containment probability gives the conditional moment, and the containment probability tends to one.

### 3.2 A checkable disk-center corollary

For the unit disk and $x = 0$, $\lambda_0(\theta) = 1/\pi^2$. With a general isotropic line intensity $c\,\mathrm{d}p\,\mathrm{d}\theta$, the measure of lines separating 0 from $y$ is $2c\|y\|$. The Poisson void formula gives

$$\mathbb{P}(y \in Z) = e^{-2c\|y\|}, \qquad \mathbb{E}\operatorname{area}(Z) = \int_{\mathbb{R}^2} e^{-2c\|y\|}\,\mathrm{d}y = \frac{\pi}{2c^2}.$$

A line-Mecke calculation sums the boundary length contributed by each candidate line; integration of its points gives

$$\mathbb{E}\operatorname{perimeter}(Z) = \pi c\,\mathbb{E}\operatorname{area}(Z) = \frac{\pi^2}{2c}.$$

These limiting-model formulas are classical, consistent with [3, 7]. The completed moment transfer supplies

$$\boxed{n^4\mathbb{E}A_n \to \frac{\pi^5}{2}, \qquad n^2\mathbb{E}P_n \to \frac{\pi^4}{2}.} \tag{7}$$

The constants refer to magnification by $n^2$. Earlier plots using $n(n-1)/\pi$ normalized the limiting length intensity to one and therefore used different numerical constants. Neither normalization should be substituted for the other.

# 4 Boundary audit: the coefficient and the missing transfer lemma

## 4.1 Exact finite geometry survives

For a hull edge placed at $(0,0)$ and $(L,0)$ with other cities $(x_i, y_i)$ above it, let

$$a = \max_i x_i/y_i, \qquad b = \max_i (L - x_i)/y_i.$$

The adjacent bounded face is the triangle with third corner $(aL/(a+b), L/(a+b))$ and area $L^2/[2(a+b)]$. The supporting rays leave every other city beyond them; connecting those cities cannot enter the triangle. This is an exact finite statement for a generic complete graph with $n \geq 4$.

The new implementation was compared to complete Shapely polygonization on 80 independently sampled small graphs ($n = 4, 7, 10, 13$, two densities). All 431 boundary triangles matched; the largest relative symmetric difference was $1.17 \times 10^{-13}$. This is numerical implementation validation, not a substitute for the supporting-ray proof.

## 4.2 The parabolic coefficient is not yet the full theorem

The prior report defines a parabolic empty-cone membership probability $p_\partial(t)$ and a convergent integral

$$c_B = \int_0^\infty p_\partial(t)\,\mathrm{d}t = 0.09169463796\ldots.$$

This limiting-model coefficient is well specified. The decimal is a quadrature evaluation, not an exact symbolic rational constant.

The missing implication is a uniform passage from the finite-city empty-cone integral to that limiting integral. Compact-window convergence alone does not justify integrating over all query depths, endpoint positions, and ray slopes. A completed proof should supply a uniform integrable envelope or a truncation argument showing that the omitted depth and slope tails vanish at the claimed $n^{-2/3}$ order. Parabolic random-hull stabilization is a relevant classical framework [4], but the complete-graph adjacent-face functional must meet its hypotheses; that is not automatic.

Accordingly, the proposed asymptotic for total boundary-face area $B_n$ stays conditional in this audit. Numerical transfer can falsify it or strengthen evidence, but cannot replace the missing estimate.

## 4.3 The density correction has a classical component

Let $f$ be a probability density positive and continuous near the boundary of a smooth positively curved convex domain. Put

$$J(f, D) = \int_{\partial D} \kappa(s)^{1/3} f(s)^{-2/3}\,\mathrm{d}s.$$

For the geometric missing area $M_n = \text{area}(D \setminus \text{conv}\{X_i\})$, the theorem of Böröczky–Fodor–Hug [5, Theorem 3.1], also stated in [6, Theorem 1.2], gives

$$\mathbb{E}M_n \sim c_M J(f, D) n^{-2/3}, \qquad c_M = (2/3)^{1/3}\Gamma(5/3).$$

This is established prior work, not a new result of this investigation.

The proposed extension for the distinct *inside-hull triangle functional* is

$$\mathbb{E}B_n \sim c_B J(f, D) n^{-2/3}.$$

It predicts a ratio of expectations

$$\frac{\mathbb{E}B_n}{\mathbb{E}M_n} \to \frac{c_B}{c_M} = 0.1162721743\ldots.$$

Applying the same geometric factor to $B_n$ is precisely the part requiring its own justification. A ratio of expectations is not an almost-sure ratio or a ratio of individual graph areas.

# 5 A frozen nonuniform-density test

## 5.1 Design fixed before generating the production data

On the unit disk, use

$$f_\eta(x, y) = \frac{1 + \eta x}{\pi}, \qquad \eta \in \{0, 0.8\}.$$

The tilted case has positive boundary density ranging from $0.2/\pi$ to $1.8/\pi$, a 9:1 contrast. Exact iid rejection sampling implements this density; there is no minimum-separation rejection.

The frozen plan specified $n = 2048, 8192$, 400 independent graphs per $(\eta, n)$, a new seed sequence, and the coefficients from the existing integral. No coefficient was re-estimated. The predicted factors are

| Density | $n^{2/3}\mathbb{E}B_n$ limit | $n^{2/3}\mathbb{E}M_n$ limit |
|---|---|---|
| Uniform | 1.235825 | 10.628727 |
| Tilted ($\eta = 0.8$) | 1.627984 | 14.001494 |

The same ratio $0.116272\ldots$ is predicted in both cases. The plan and production-code hash were saved before the production calls. This was an internal computational precommitment, not registration with an external registry.

## 5.2 Results, including the finite-size miss

The production test constructed 86,686 boundary triangles across 1,600 graphs. The table uses 5,000 whole-graph bootstrap resamples per group with a fixed bootstrap seed. Intervals are pointwise, not simultaneous.

| Density | $n$ | $n^{2/3}\overline{B}$ | 95% interval | Frozen limit |
|---|---|---|---|---|
| Uniform | 2,048 | 1.24069 | [1.21952,1.26144] | 1.23583 |
| Uniform | 8,192 | 1.23740 | [1.22128,1.25400] | 1.23583 |
| Tilted | 2,048 | 1.56575 | [1.53388,1.59888] | 1.62798 |
| Tilted | 8,192 | 1.60864 | [1.58097,1.63620] | 1.62798 |

The tilted 2,048-city result is about 3.8% below its limiting prediction, outside its Monte Carlo interval. Its missing-area result also lies below its asymptotic prediction. At 8,192 cities both respective predictions are within the pointwise intervals. These are finite-size observations, not proof that the remaining bias has a specified rate.

| Density | $n$ | Ratio of sample means $\overline{B}/\overline{M}$ | 95% interval |
|---|---|---|---|
| Uniform | 2,048 | 0.116750 | [0.114176,0.119315] |
| Uniform | 8,192 | 0.116447 | [0.114714,0.118310] |
| Tilted | 2,048 | 0.114985 | [0.111885,0.118234] |
| Tilted | 8,192 | 0.115683 | [0.113339,0.118143] |

All four ratio intervals contain the frozen target. The cancellation is supported by this density perturbation, but these four intervals do not establish a density-universal theorem. No additional sample sizes were selected after seeing the initial results.

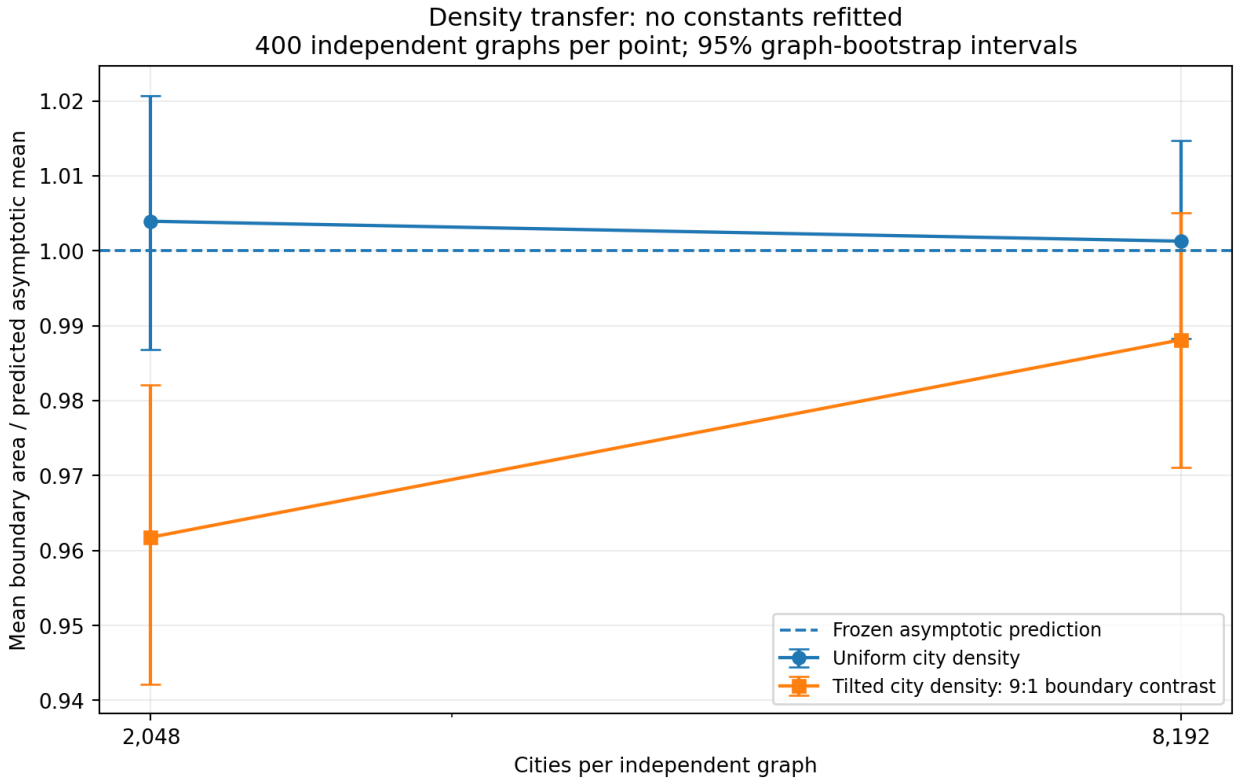



Figure 1: Observed boundary-face means normalized by the frozen asymptotic predictions. The tilted finite-size bias is visible rather than absorbed in a fitted coefficient.

# 6 Fan audit: exact algebra is not process identification

## 6.1 What is exact, and what remains conjectural

The prior proposed fan field starts with a unit Poisson cloud of sources $(X, T) \in \mathbb{R} \times (0, 1)$. Given the sources, a source contributes parallel lines

$$v + (T/X)u = c$$

with offset intensity $T^2|X|^{-3}\,\mathrm{d}c$. This defines a locally finite random line arrangement almost surely, although its unconditional mean line intensity is infinite.

The normal-ray rate is $W_3 = \sum T^2/|X|^3$ and the tangent-ray rate is $W_4 = \sum T^3/|X|^4$. The Poisson Laplace functional [7, Theorem 3.9] gives the exact model identities

$$\begin{aligned} \mathbb{E}e^{-sW_3} &= e^{-c_3 s^{1/3}}, & c_3 &= \tfrac{6}{5}\Gamma(2/3), \\ \mathbb{E}e^{-tW_4} &= e^{-c_4 t^{1/4}}, & c_4 &= \tfrac{8}{7}\Gamma(3/4). \end{aligned}$$

The Cox-process definition and positive-stable Laplace forms are classical probability tools. The geometric source weights may be a useful complete-graph derivation, but the Laplace algebra is not itself a new general stochastic-process result.

The exact finite-binomial conditional estimator also survives audit: given all cap cities, the remaining cities are independent outside the cap, and avoidance of a forbidden endpoint union $U$ has probability $(1 - |U|/|D \setminus \mathrm{cap}|)^{n-K}$ unless a known cap–cap segment already hits. The identity is exact; the implementation's union areas are numerical quadrature quantities.

The full limit from original graphs requires more: conditional joint convergence of remote-endpoint wedge counts; negligible overlaps shared by multiple source cities; uniform control when $X \to 0$, $T \to 0$, and $|X| \to \infty$; and a justified order of the coupled limits $M \to \infty$, $\tau \to 0$. These conditions are not supplied by agreement of a finite list of survival curves.

### 6.2 An explicit rival geometry passes the previous ray tests

Let $\widetilde{W}_3$ and $\widetilde{W}_4$ be *independent* positive random variables with the two Laplace transforms above. Conditional on these variables, draw horizontal Poisson lines at rate $\widetilde{W}_3$ and vertical Poisson lines at rate $\widetilde{W}_4$.

This is a stationary rectangular Cox grid. It has exactly the same normal-ray and tangent-ray survival functions as the fan model. It also gives, for adjacent collinear normal intervals of lengths $s, t$,

$$\mathbb{P}(\text{both empty}) = e^{-c_3(s+t)^{1/3}},$$

so it matches the previously emphasized adjacent-normal-gap dependence as well. But every cell is a rectangle, and its two directional rate variables are independent. It is not the proposed fan geometry.

This is an identifiability counterexample, not a claim that the grid is the correct city-generated model or that the directly observed fan picture never occurs. It proves that those chosen probability checks alone do not identify the full two-dimensional limiting process.

### 6.3 A separating joint event with no new fitted parameters

Take a positive normal unit ray and a positive tangent unit ray from the same origin, in the previously derived anisotropic coordinates. The rival grid predicts

$$\mathbb{P}(\text{both rays empty}) = e^{-c_3 - c_4} = 0.04853754\ldots. \tag{8}$$

For a fan source with $X = x > 0$, the offset intervals overlap, so the union length is $\max(s, Tt/x)$. For $X = -x < 0$, they lie on opposite sides and the union length is $s + Tt/x$. Thus the fan prediction has exponent

$$\begin{aligned} \Psi_{NT}(s,t) = \int_0^1 \int_0^\infty \Big[ & 1 - e^{-T^2 x^{-3} \max(s, Tt/x)} \\ & + 1 - e^{-T^2 x^{-3}(s + Tt/x)} \Big] \,\mathrm{d}x\,\mathrm{d}T. \end{aligned} \tag{9}$$

At $s = t = 1$, evaluation gives

$$\boxed{e^{-\Psi_{NT}(1,1)} = 0.17400595\ldots,} \tag{10}$$

over 3.5 times the grid prediction. It is incorrect to treat both source signs as a simple sum of ray rates: the same source's line can hit both rays, and the overlap must be accounted for.

For stable quadrature, put $x = Tu$ and define $E_3(a) = \int_1^\infty e^{-ay} y^{-3}\,\mathrm{d}y$. Then

$$\Psi_{NT}(1,1) = \int_0^\infty \left[1 - E_3\left(\frac{\max(1,1/u)}{u^3}\right) - E_3\left(\frac{1+1/u}{u^3}\right)\right] \mathrm{d}u.$$

The numerical code recovers $c_3$ and $c_4$ when one ray length is set to zero, to absolute discrepancies below $1.1 \times 10^{-13}$. The quadrature's own reported error is a numerical diagnostic, not a rigorous interval proof.

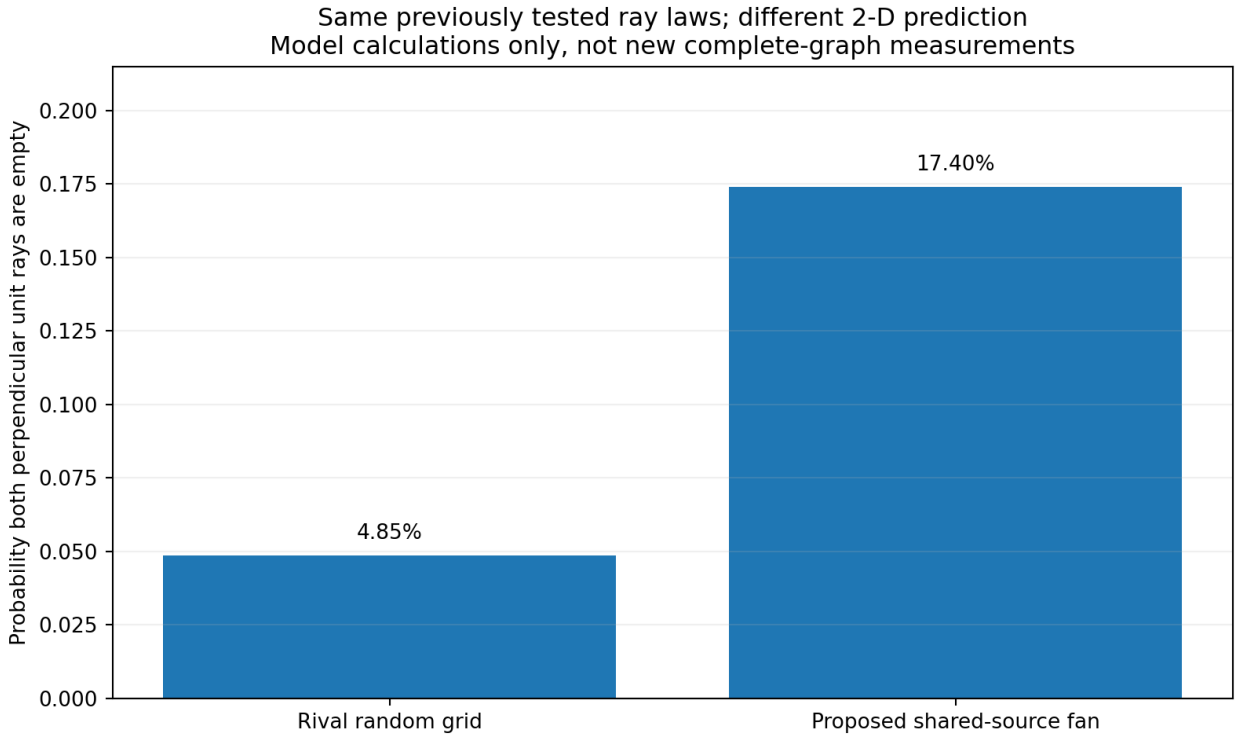



Figure 2: A predeclared discriminator between two limiting models. These are computed model probabilities, not new complete-graph observations. The finite-graph perpendicular-ray experiment was not run in this audit.

This is a much more informative next validation target than increasing the sample size on a marginal curve that both geometries reproduce. Even success would remain evidence for the conjecture, not a substitute for the joint-convergence proof.

# 7 Literature and priority: the corrected map

## 7.1 Established results that must be credited

**Functional Poisson approximation.** Decreusefond–Schulte–Thäle [1] explicitly treat point processes obtained by symmetric maps of all $k$-tuples of binomial or Poisson input. Their binomial order-two

overlap term is the appropriate engine for the bulk limit. Related extreme-value convergence criteria are surveyed in [2].

**Zero-cell laws.** Calka's conditional-perimeter result [3, Corollary 2] includes an explicit remark crediting Matheron for the Gamma law. The classical side-count and zero-versus-typical-cell distinctions are not new discoveries from our simulations. Their value here is as predictions for a different input-generated model after its limit has been proved.

**Parabolic random-hull scaling.** The $n^{-1/3}$ tangential and $n^{-2/3}$ normal scales belong to the established random-polytope framework; Calka–Schreiber–Yukich [4] provide relevant local-limit and stabilization theory. New complete-graph boundary functionals require their own verified transfer.

**Weighted missing-area formula.** Böröczky–Fodor–Hug [5, Theorem 3.1] prove the $f^{-2/3}\kappa^{1/3}$ factor in dimension two. Besau–Ludwig–Werner [6] reproduce the theorem and its constant. Finding that correction here is a connection to existing work, not a novelty claim.

**Cox and shot-noise machinery.** Last–Penrose [7, Chapters 3 and 13] provide the Poisson Laplace functional and random-intensity framework. Once inverse-power source weights are assumed, the stable exponents follow from that established calculus. The unproved issue is whether the full random-city graph converges to that particular directing measure.

### 7.2 What the search can and cannot establish

The audit used public-web searches for the actual construction, including complete geometric graphs, all-pair segments, local Poisson lines, hull-edge triangular faces, and the boundary/fan formulas. It inspected the relevant theorem statements and followed primary references, rather than relying only on abstracts.

No directly matching publication establishing the exact three-part complete-graph package was identified among the inspected sources. That is not a novelty certificate. The search was not an exhaustive MathSciNet or zbMATH citation-index review, and terminology can hide an equivalent result.

One recurring search hit, [8], converts an existing tessellation into a colored complete graph on its cells. That is the reverse construction, not the subdivision made by drawing all segments between sampled cities. It should not be cited as either proof of priority or proof of difference beyond that explicit model comparison.

The bibliography supports a narrower positioning: a specific geometric application and moment argument, a separate boundary functional with a candidate coefficient, and a proposed intermediate model. No result here is labeled "first" or assigned a numerical novelty probability.

## 8 A manuscript-ready structure and stopping rules

The best-supported core is a paper provisionally titled *Local cells of complete geometric graphs: Poisson limits and moment bounds*. Its main content would be Theorem 1, the explicit angular intensity, exact local mean fields, and a compact independent validation section. Classical zero-cell formulas should be cited as corollaries of the identified limit, not sold as discoveries.

The boundary coefficient is a potential second contribution only after the localization and tail estimates are written in a form that an external reader can check. The completed weighted missing-area literature theorem can be used directly; the complete-graph triangle functional cannot simply inherit it by analogy.

The fan regime should remain a clearly marked conjectural section or separate project until the joint

wedge and truncation estimates are completed. The perpendicular-ray event above is a predeclared model discriminator. It has more information value than another marginal survival sweep, but does not become a theorem if it agrees.

The concrete completion criteria are therefore: independent verification of the fixed-interior proof and constants; a source-by-source priority check; an integrable boundary-localization bound; and conditional joint convergence for a finite family of genuinely two-dimensional test windows in the fan regime. None is replaced by adding more decimal places to a fit.

## 9 Reproducibility and scope of the supplied files

The archive contains the production density sampler, exact boundary-triangle implementation, frozen plan, all 1,600 per-graph records, independent polygonization checks, the rival-model joint integral, figure script, environment metadata, claim ledger, source matrix, search log, and this editable report.

The canonical new summary, `data/summary_recomputed.csv`, is produced by the script `code/validate_and_summarize.py`. The initial summary is retained as provenance; its point estimates are identical and its Monte Carlo bootstrap endpoint differences are inconsequential. The report uses the fully scripted reconstruction. The source sampler is unchanged from its frozen hash.

The bundle records hashes of the three earlier report sources but does not copy the old archives or third-party papers. No prior simulation totals are added to the new totals. The two-direction joint-void values are model calculations only, and this audit contains no new complete-graph fan-run data.

**Bottom line.** The audit yields one stronger theorem-level argument, one density-transfer experiment with an honest finite-size miss, and one explicit countermodel to an overstrong inference. That is consolidation rather than another escalation of loosely connected claims.

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